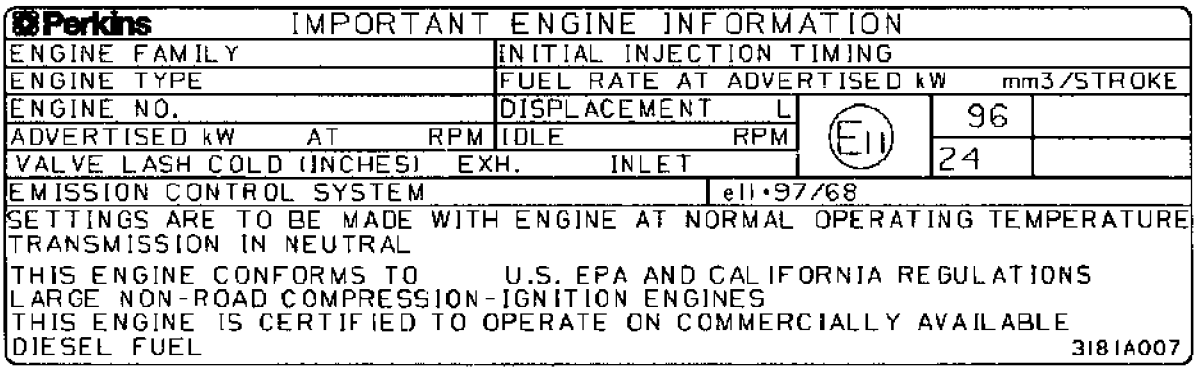
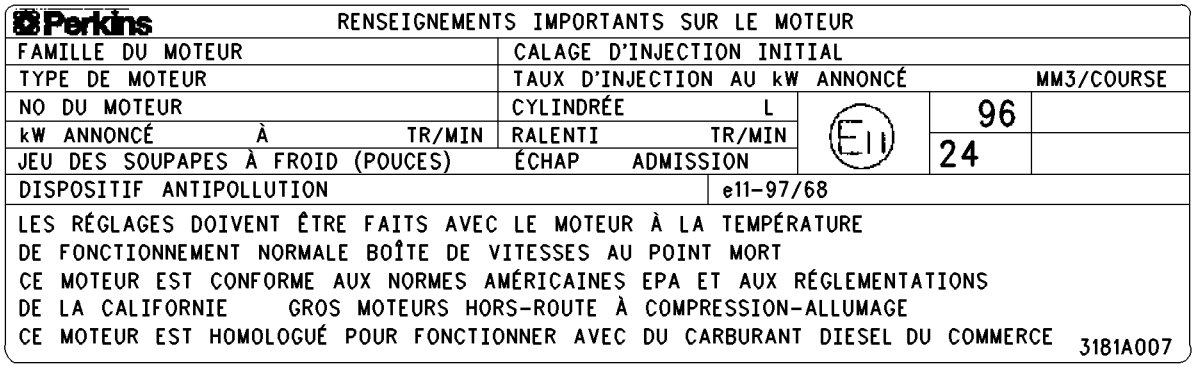


Perkins	IMPORTANT ENGINE INFORMATION				
ENGINE FAMILY		INITIAL INJECTION TIMING			
ENGINE TYPE		FUEL RATE AT ADVERTISED kW			mm3/STROKE
ENGINE NO.		DISPLACEMENT L	E11	96	
ADVERTISED kW AT RPM		IDLE RPM		24	
VALVE LASH COLD (INCHES) EXH.	INLET				
EMISSION CONTROL SYSTEM		e11•97/68			
SETTINGS ARE TO BE MADE WITH ENGINE AT NORMAL OPERATING TEMPERATURE TRANSMISSION IN NEUTRAL					
THIS ENGINE CONFORMS TO U.S. EPA AND CALIFORNIA REGULATIONS LARGE NON-ROAD COMPRESSION-IGNITION ENGINES THIS ENGINE IS CERTIFIED TO OPERATE ON COMMERCIALLY AVAILABLE DIESEL FUEL					3181A007

The EPA/EU Emissions Certification Film (if applicable) is located either on the side, the top, or the front of the engine.



Perkins	RENSEIGNEMENTS IMPORTANTS SUR LE MOTEUR				
FAMILLE DU MOTEUR		CALAGE D'INJECTION INITIAL			
TYPE DE MOTEUR		TAUX D'INJECTION AU kW ANNONCÉ			MM3/COURSE
NO DU MOTEUR		CYLINDRÉE L	E11	96	
kW ANNONCÉ À TR/MIN		RALENTI TR/MIN		24	
JEU DES SOUPAPES À FROID (POUCES)		ÉCHAP ADMISSION			
DISPOSITIF ANTIPOLLUTION		e11-97/68			
LES RÉGLAGES DOIVENT ÊTRE FAITS AVEC LE MOTEUR À LA TEMPÉRATURE DE FONCTIONNEMENT NORMALE BOÎTE DE VITESSES AU POINT MORT CE MOTEUR EST CONFORME AUX NORMES AMÉRICAINES EPA ET AUX RÉGLEMENTATIONS DE LA CALIFORNIE GROS MOTEURS HORS-ROUTE À COMPRESSION-ALLUMAGE CE MOTEUR EST HOMOLOGUÉ POUR FONCTIONNER AVEC DU CARBURANT DIESEL DU COMMERCE					3181A007

L'AUTOCOLLANT D'HOMOLOGATION DU DISPOSITIF ANTIPOLLUTION EPA/UE (SELON ÉQUIPEMENT) EST SITUÉ SOIT SUR LE CÔTÉ, SOIT SUR LE DESSUS DU MOTEUR SOIT SUR LE DEVANT DU MOTEUR.

Illustration 23 g00937288

Operation Section

Before Operation

i04027420

Mounting and Dismounting

SMCS Code: 7000

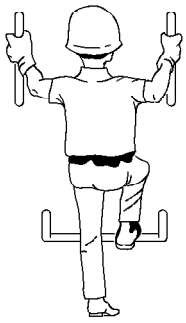


Illustration 24 g00037860

Use steps and handholds whenever you mount the machine. Use steps and handholds whenever you dismount the machine. Before you mount the machine, clean the step and the handholds. Inspect the step and handholds. Make all necessary repairs.

Face the machine whenever you mount the machine and whenever you dismount the machine. Maintain a three-point contact with the step and with handholds.

Note: Three-point contact can be two feet and one hand. Three-point contact can also be one foot and two hands.

Do not mount a moving machine. Do not dismount a moving machine. Never jump off the machine. Do not try to mount the machine when you carry tools or supplies. Do not try to dismount the machine when you are carrying tools or supplies. Use a hand line to pull equipment onto the platform. Do not use any controls as handholds when you enter the operator compartment or when you exit the operator compartment.

Machine Access System Specifications

The machine access system has been designed to meet the intent of the technical requirements in “ISO 2867 Earth-moving Machinery – Access Systems”. The access system provides for operator access to the operator station and to conduct the maintenance procedures described in Maintenance section.

i01977672

Daily Inspection

SMCS Code: 1000; 7000

For a maximum service life of the machine, complete a thorough walk-around inspection before you mount the machine and before you start the engine.

Inspect the area around the machine and under the machine. Look for loose bolts, trash buildup, oil, coolant leakage, broken parts, or worn parts.

Note: Watch closely for leaks. If you observe a leak, find the source of the leak and correct the leak. If you suspect a leak or you observe a leak, check the fluid levels more frequently.

Inspect the condition of the equipment and of the hydraulic components.

Check the condition of the tires. Adjust the inflation pressure, if necessary.

Check all of the oil levels, all of the coolant levels, and all of the fuel levels.

Remove any trash buildup and debris. Make all necessary repairs before you operate the machine.

Make sure that all covers and guards are securely attached.

Adjust the mirrors for the correct rear view of the machine.

Make sure that the engine air filter service indicator is not in the red zone.

Grease all of the fittings that need to be serviced on a daily basis.

Daily, perform the procedures that are applicable to your machine:

- Backup Alarm - Test
- Cooling System Level - Check
- Drum Scrapers - Inspect/Adjust/Replace
- Engine Air Filter Service Indicator - Inspect
- Engine Oil Level - Check

- Hydraulic System Oil Level - Check
- Indicators and Gauges - Test
- Neutral Start Switch - Test
- Seat Belt - Inspect
- Tire Scraper - Inspect/Adjust/Replace
- Water Spray System Filter - Clean

i01977753

Steering Frame Lock

SMCS Code: 7506

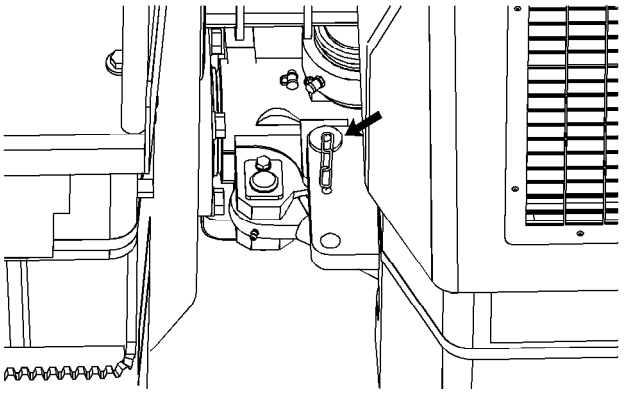


Illustration 25 g01031425

The steering frame lock is in the LOCKED position.

Install the steering frame lock link in the LOCKED position before you lift the machine and before you transport the machine on another vehicle. Also install the steering frame lock link before you perform maintenance near the center of the machine.

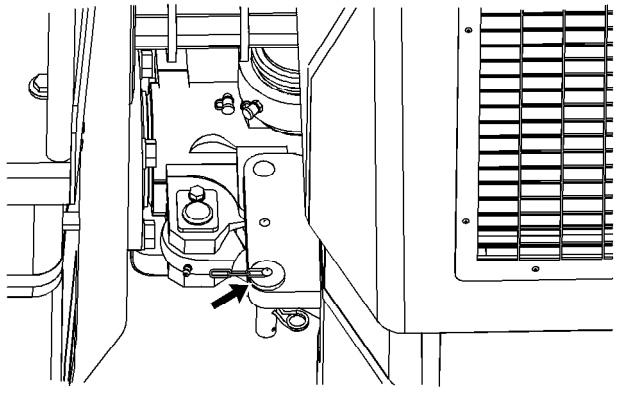


Illustration 26 g01041171

The steering frame lock is in the UNLOCKED position.

Install the steering frame lock in the UNLOCKED position before you operate the machine.

Machine Operation

i02021431

Seat

SMCS Code: 7312; 7324

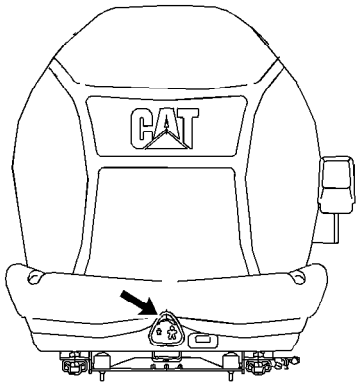


Illustration 27 g01043870

Note: Adjust the seat at the beginning of every shift. Adjust the seat when operators change.

When the operator's back is against the back of the seat, adjust the seat so that the controls can be reached.

Note: Always lock the seat in position before you operate the machine.

In order to adjust the suspension of the seat, turn the knob. Turn the knob clockwise in order to increase the stiffness of the suspension of the seat. Turn the knob counterclockwise in order to decrease the stiffness of the suspension of the seat.

Sliding Seat (If Equipped)

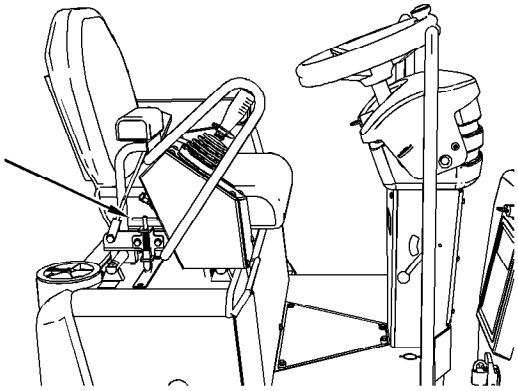


Illustration 28 g01043871

Lift the pin and slide the seat assembly to the RIGHT, CENTER, or LEFT position. Release the pin in order to lock the seat in the desired position.

Note: Always lock the seat in position before you operate the machine.

i04200349

Seat Belt

SMCS Code: 7327

Note: This machine was equipped with a seat belt when the machine was shipped from Caterpillar. At the time of installation, the seat belt and the instructions for installation of the seat belt meet the SAE J386 and ISO 6683 standards. Consult your Cat dealer for all replacement parts.

Always check the condition of the seat belt and the condition of the mounting hardware before you operate the machine.

Seat Belt Adjustment for Non-Retractable Seat Belts

Adjust both ends of the seat belt. The seat belt should be snug but comfortable.

Lengthening the Seat Belt

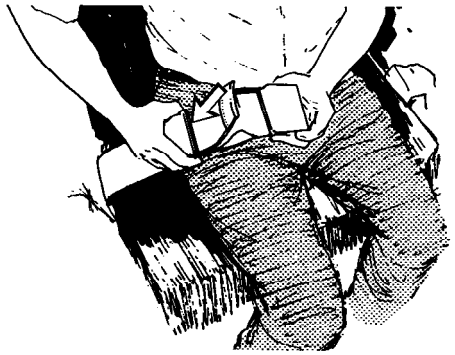


Illustration 29 g00100709

1. Unfasten the seat belt.

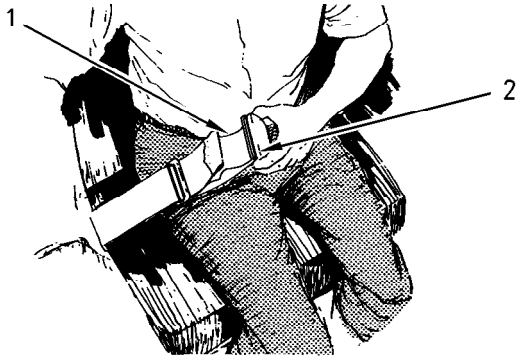


Illustration 30 g00932817

2. To remove the slack in outer loop (1), rotate buckle (2). This will free the lock bar. This permits the seat belt to move through the buckle.

3. Remove the slack from the outer belt loop by pulling on the buckle.

4. Loosen the other half of the seat belt in the same manner. If the seat belt does not fit snugly with the buckle in the center, readjust the seat belt.

Shortening the Seat Belt



Illustration 31 g00100713

1. Fasten the seat belt. Pull out on the outer belt loop in order to tighten the seat belt.

2. Adjust the other half of the seat belt in the same manner.

3. If the seat belt does not fit snugly with the buckle in the center, readjust the seat belt.

Fastening The Seat Belt

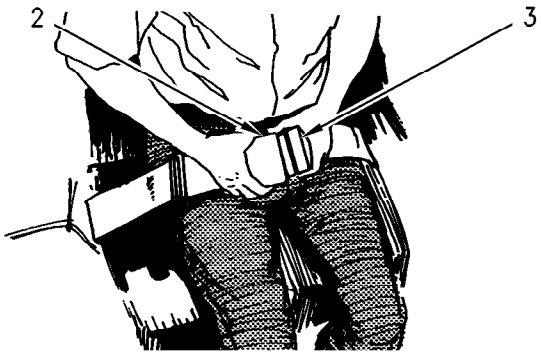


Illustration 32 g00932818

Fasten the seat belt catch (3) into the buckle (2). Make sure that the seat belt is placed low across the lap of the operator.

Releasing The Seat Belt



Illustration 33 g00100717

Pull up on the release lever. This will release the seat belt.

Seat Belt Adjustment for Retractable Seat Belts

Fastening The Seat Belt

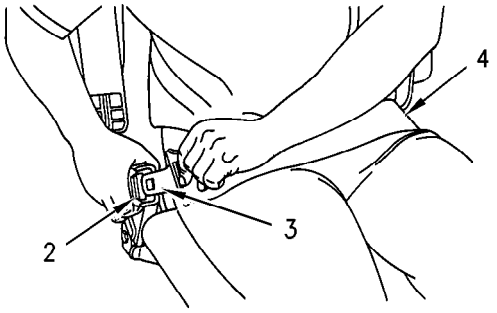


Illustration 34 g00867598

Pull seat belt (4) out of the retractor in a continuous motion.

Fasten seat belt catch (3) into buckle (2). Make sure that the seat belt is placed low across the lap of the operator.

The retractor will adjust the belt length and the retractor will lock in place. The comfort ride sleeve will allow the operator to have limited movement.

Releasing The Seat Belt

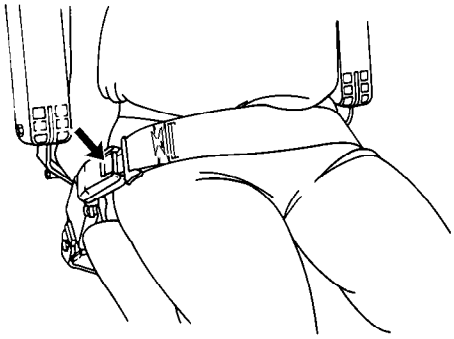


Illustration 35 g00039113

Push the release button on the buckle in order to release the seat belt. The seat belt will automatically retract into the retractor.

Extension of the Seat Belt

⚠ WARNING

When using retractable seat belts, do not use seat belt extensions, or personal injury or death can result.

The retractor system may or may not lock up depending on the length of the extension and the size of the person. If the retractor does not lock up, the seat belt will not retain the person.

Longer, non-retractable seat belts and extensions for the non-retractable seat belts are available.

Caterpillar requires only non-retractable seat belts to be used with a seat belt extension.

Consult your Cat dealer for longer seat belts and for information on extending the seat belts.

i03395700

Operator Controls

SMCS Code: 7000; 7300; 7301; 7451

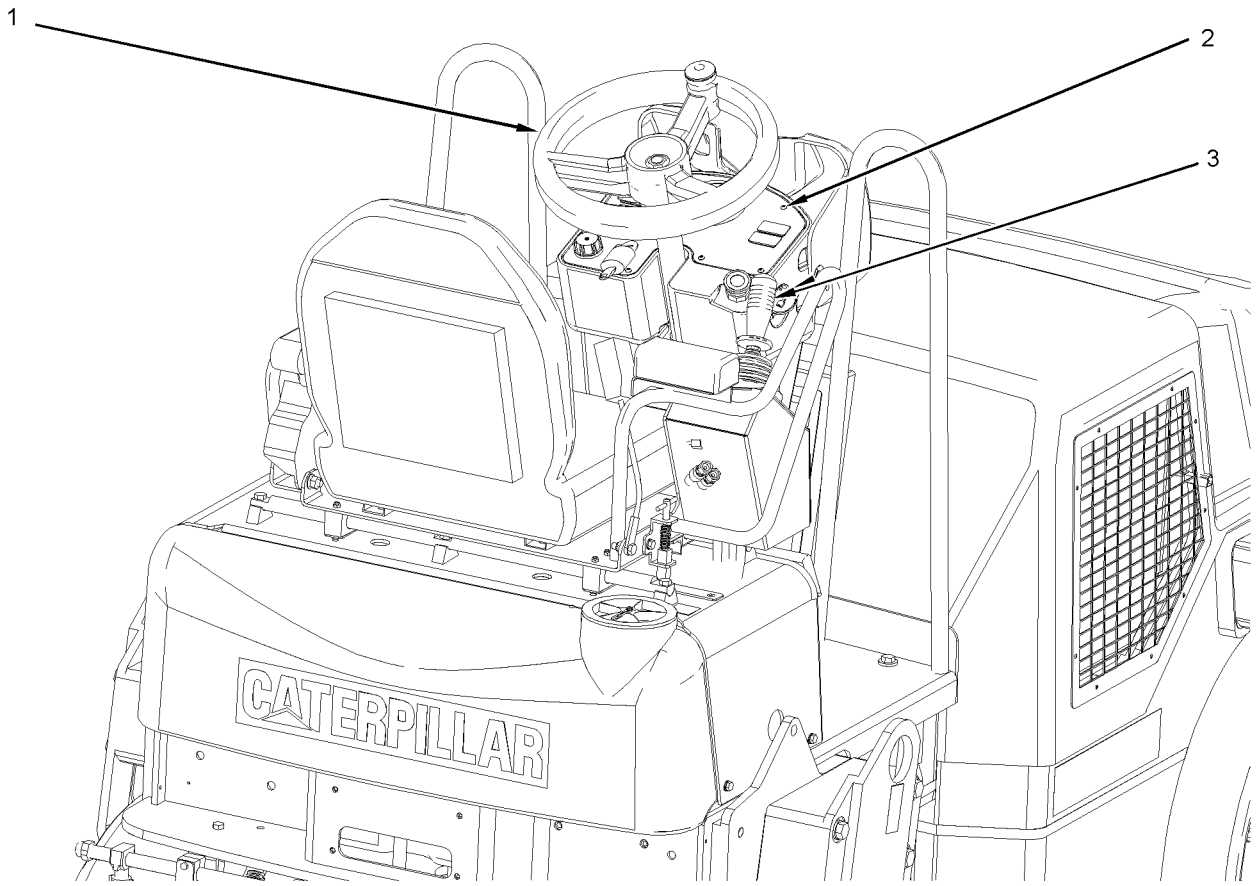


Illustration 36 g01030787

(1) Steering wheel (2) Steering console (3) Propel control lever

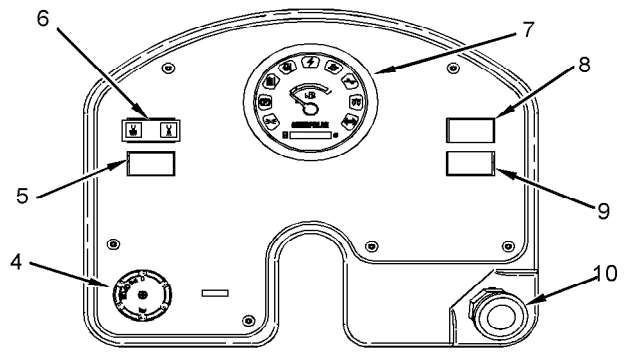


Illustration 37 g01030790

Close up view of steering console (2)

(4) Switch for the light and the horn
(5) Vibratory control switch CB-214E and CB-224E
(5) Additive Spray Switch CB-225E
(6) Switch for the water spray
(7) Alert indicators
(8) Switch for the turn signal
(9) Switch for the hazard lights
(10) Parking brake control

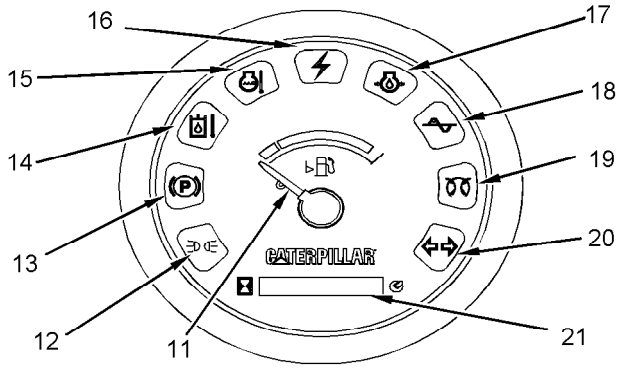


Illustration 38 g01031570

Close up view of the fuel gauge and alert indicators (7)

(11) Indicator for the fuel level
(12) Indicator for the roading lights
(13) Indicator for the parking brake
(14) Indicator for the hydraulic oil temperature
(15) Indicator for the engine coolant temperature
(16) Indicator for the alternator
(17) Indicator for the engine oil pressure
(18) Indicator for the vibratory system
(19) Indicator for the engine preheater
(20) Indicator for the turn signals
(21) Service hour meter

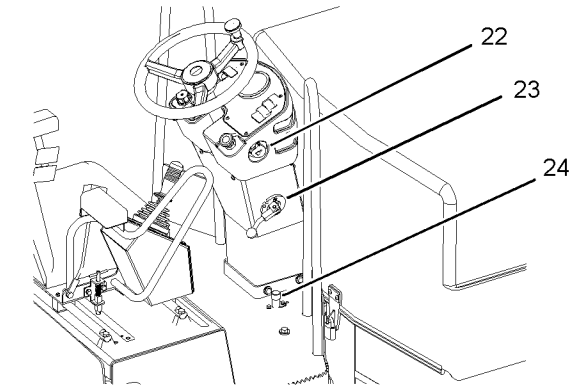


Illustration 39 g01415258

(22) Throttle Control
(23) Engine Start Switch
(24) Traction Control Switch

Propel Control Lever (3)

FORWARD – In order to move the machine forward, slowly push the propel control lever away from the center NEUTRAL position.

NEUTRAL – Move the propel control lever to the NEUTRAL position in order to stop the machine.

REVERSE – In order to move the machine in reverse, slowly pull the propel control lever away from the center NEUTRAL position.

On/Off Switch for the Vibratory System

The switch on the propel lever controls the operation of the vibration system.

ON – In order to turn on the vibration system, press the switch.

OFF – In order to stop the vibration system, press the switch again.

Switch for the Lights and the Horn (4)

OFF – Turn the knob counterclockwise in order to turn off all of the lights.

Parking Lights – Turn the knob clockwise to the first position in order to turn on the parking lights.



Roading Lights – Turn the knob clockwise to the second position in order to turn on the roading lights.



Working Lights – Turn the knob clockwise to the third position in order to turn on the work lights.



Horn – Push the knob in order to activate the horn.

Switch for the Vibratory Control (5)

Note: This is only for CB-214E and CB-224E machines.

Standard Vibratory Control System



Vibratory Control – Move the switch to the left in order to turn on the vibratory system for the front drum and the rear drum.



Vibratory Control – Move the switch to the right in order to turn on the vibratory system for the front drum only.

Optional Vibration Control



Vibratory Control – Move the switch to the left in order to turn on the vibratory system for the rear drum only.



Vibratory Control – Move the switch to the middle position in order to turn on the vibratory system for the front drum and the rear drum.



Vibratory Control – Move the switch to the right position in order to turn on the vibratory system for the front drum only.

Additive Spray Switch (5)

Note: The emulsion spray control is equipped on the CB-225E machines only. The emulsion spray is used in order to prevent the accumulation of asphalt on the rear tires.

Move the switch to the left position in order to turn off the emulsion spray. Move the switch to the right in order to turn on the emulsion spray.

Switch for the Water Spray (6)

NOTICE
Always check the water level in the water tanks before you use the water spray system.

Contaminated water in the water tanks can cause performance problems and reliability poroblems in the water spray sytem.

NOTICE
Drain the water spray system completely when freezing conditions exist. This is done in order to avoid damage to the water spray system.



Intermittent Spray – Move the switch to the right position in order to activate the intermittent water spray system.

OFF – Move the switch to the center position in order to turn the water spray system OFF.



Continuous Spray – Move the switch to the left position in order to activate the continuous water spray system.

Switch for the Turn Signal (8)

Turn Signal – Move the switch for the turn signal to the left in order to activate the left turn signal lights. Move the switch for the turn signal to the right in order to activate the right turn signal lights.

Switch for the Hazard Lights (9)

Hazard Lights – Move the switch to the right in order to activate the hazard lights. Move the switch to the left in order to deactivate the hazard lights.

Control for the Parking Brake (10)



Parking Brake – In order to apply the parking brake, push the knob downward. Pull the knob upward in order to release the parking brake.

Fuel Level Gauge (11)



Fuel Level Gauge – The fuel level gauge indicates the amount of diesel fuel in the fuel tank.

Indicator for the Roading Lights (12)



Roading Lights – The indicator will illuminate when the roading lights are on.

Indicator for the Parking Brake (13)



Parking Brake – The indicator will illuminate when the parking brake is applied.

Indicator for the Hydraulic Oil Temperature (14)



Hydraulic Oil Temperature – The indicator will illuminate when the hydraulic oil in the hydraulic tank is too hot.

Note: If the indicator for the hydraulic oil temperature illuminates, stop the machine. Move the throttle lever to low idle. Allow the machine to idle for several minutes. Resume using the machine if the hydraulic oil temperature returns to normal. If the indicator does not turn OFF, or if the indicator illuminates again, stop the machine and investigate the problem. Do not operate the machine until all of the repairs have been made.

Indicator for the Engine Coolant Temperature (15)



Engine Coolant Temperature – The indicator will illuminate when the temperature of the engine coolant is too hot. Stop the machine and investigate the problem. Do not operate the machine until all of the repairs have been made.

Indicator for the Charging System (16)



Charging System – The indicator will illuminate when the alternator is malfunctioning. If the indicator illuminates, move the machine to a safe area and stop the machine. Investigate the problem. Do not operate the machine until all of the repairs have been made.

Indicator for the Engine Oil Pressure (17)



Engine Oil Pressure – The indicator will illuminate when the engine oil pressure is too low. If the indicator illuminates, stop the engine and investigate the problem. Do not operate the machine until all of the repairs have been made.

Indicator for the Vibratory System (18)



Vibratory System – The indicator will illuminate when the vibratory system is activated.

Indicator for the Engine Preheater (19)



Engine Preheater – The indicator will illuminate when the engine start switch is turned to the PREHEAT position. When the indicator turns off, turn the engine start switch to the START position in order to start the engine.

Indicator for the Turn Signals (20)

Turn Signals – The indicator will illuminate when the turn signals are activated.

Service Hour Meter (21)

Service Hour Meter – This gauge indicates the total operating hours of the engine. To determine the service hour maintenance intervals, use the service hour meter.

Throttle Control (22)

LOW – In order to decrease the engine speed to low speed, move the throttle control downward to the LOW position.

HIGH – In order to increase the engine speed to high speed, move the throttle control upward to the HIGH position.

Engine Start Switch (23)

OFF – In order to disconnect the electrical power to the engine and to the machine, turn the switch in a counterclockwise direction to the OFF position. Turn the switch to the OFF position before trying to restart the engine. Turn the switch to the OFF position in order to stop the engine.

ON – In order to activate the machine circuits, turn the switch to the ON position. When the switch is released from the START position, the switch will return to the ON position.

START – Turn the engine start switch to the START position in order to crank the engine. Release the key as soon as the engine starts.

Note: If the engine does not start, return the switch to the OFF position before returning to the START position.

For more information, refer to the Operation and Maintenance Manual, “Engine Starting” section.

Traction Control Switch (24) (If Equipped)

The traction control switch is located on the floor to the right of the steering console. The traction control switch controls the traction control system. If the front drum or the rear drum begins to slip, depress the traction control switch in order to limit the slipping of the drum.

Note: When you operate the traction control system reduce the propel speed until the drum stops spinning. Excessive movement of the propel lever may cause the engine to lug when the traction control system is engaged.

i01994352

Operation Information

SMCS Code: 1000; 7000

Operate the machine only when the operator is correctly seated in the operator's compartment.

Operate the machine with a secured seat belt.

Operate the controls only when the engine is operating.

Inspect the machine for proper operation of controls.

Operate the machine slowly in an open area in order to inspect the protective devices.

The work area must be clear. The operator can then proceed to operate the machine.

Only the operator is allowed on the machine.

Report any needed repairs.

Maintain a safe distance from the edge of cliffs, overhangs and slide areas.

i02013920

Backup Alarm

SMCS Code: 7406

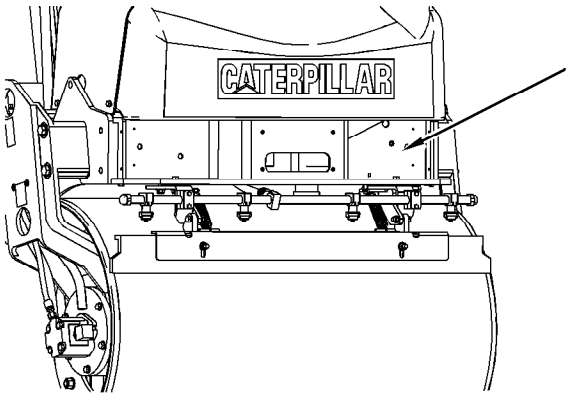


Illustration 40 g01041221

The backup alarm is located on the rear of the machine.

Backup Alarm – The backup alarm will sound when the propel lever is in the REVERSE position. The backup alarm alerts any personnel that the machine is backing up.

i02202400

Rollover Protective Structure (ROPS)
(Foldable)

SMCS Code: 7323

If your machine is equipped with a foldable rollover protective structure (ROPS), the ROPS must be in the raised position before you operate the machine. Before the machine is shipped, lower the ROPS in order to provide more clearance during transportation.

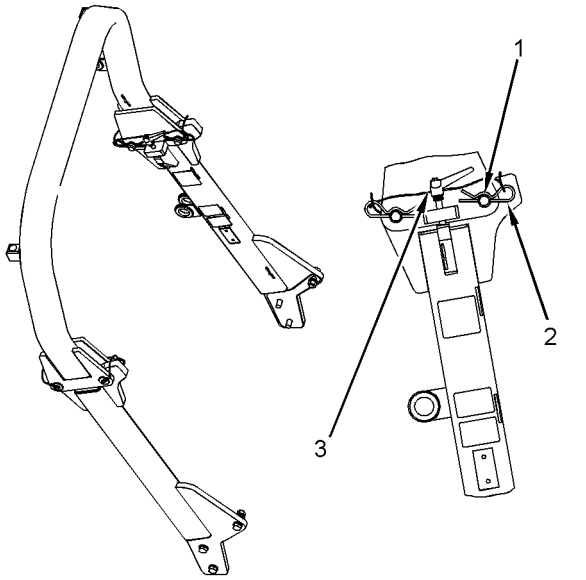


Illustration 41 g01112085

Raise

In order to raise the ROPS, perform the following steps:

1. Raise the ROPS to the full upright position.

2. Install front locking pins (1) through the ROPS post.

3. Install front cotter pins (2) through front locking pins (1). This will lock the locking pins in place.

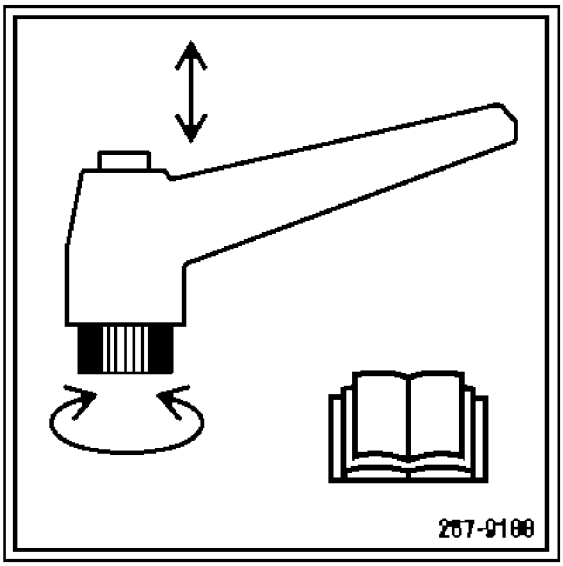


Illustration 42 g01114409

4. Tighten locking lever (3).

Note: In order to tighten the locking lever, perform the following procedure:

a. Turn the locking lever clockwise. Turn the locking lever by one-half turn.

b. Raise the locking lever.

c. Turn the locking lever counterclockwise by one-half turn.

d. Release the locking lever. This will allow the locking lever to lower.

e. Repeat step 4a and step 4d until the locking lever is tight. Move the locking lever to the full counterclockwise position after the locking lever is tightened.

Lower

In order to lower the ROPS, perform the following steps:

1. Loosen locking lever (3).

Note: In order to loosen the locking lever, perform the following procedure:

a. Raise the locking lever.

b. Turn the locking lever clockwise by one-half turn.

c. Release the locking lever. This will allow the locking lever to lower.

d. Turn the locking lever counterclockwise by one-half turn.

e. Repeat step 1a and 1d until the locking lever is loosened.

2. Remove front cotter pins (2).

3. Remove front locking pins (1).

4. Lower the ROPS toward the rear of the machine.

5. In order to store the front locking pins, install the front locking pins into the holes on the ROPS post.

6. Install front cotter pins (2) through front locking pins (1). This will lock the locking pins in place.

Engine Starting

i01977827

Engine Starting

SMCS Code: 1000; 7000

⚠ WARNING

To prevent personal injury, be sure all personnel are clear of the machine. Sound the horn for several seconds before starting the engine.

⚠ WARNING

Do not use ether. This machine is equipped with glow plugs. Using ether can create explosions or fires that can cause personal injury or death. Read and follow the engine starting procedure in the Operation and Maintenance Manual.

NOTICE
Do not crank the engine for more than 30 seconds. Allow the starter to cool for two minutes before cranking again.

The key must be in run position with the engine running to maintain electrical and hydraulic functions and prevent machine damage.

If the engine does not start, return the key to OFF before returning it to START.

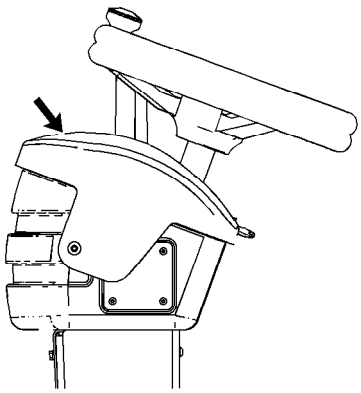


Illustration 43 g01041265

The vandalism guard is in the CLOSED position.

1. Open the vandalism guard.

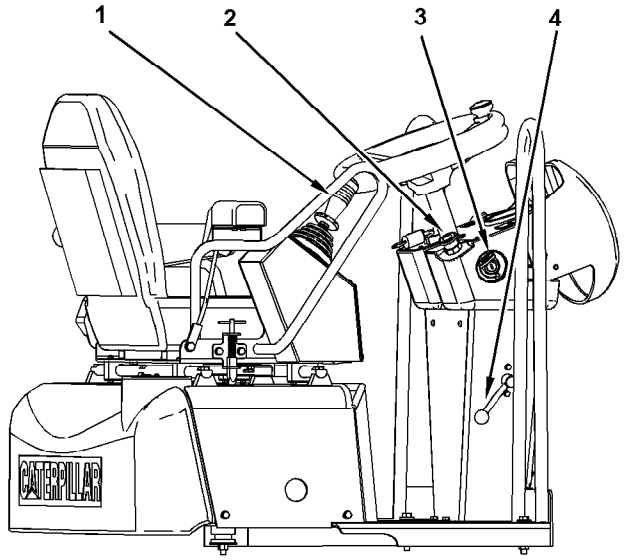


Illustration 44 g01032136

2. Move propel control lever (1) to the NEUTRAL position.

3. Push parking brake knob (2) downward in order to engage the parking brake.

4. Move throttle control (4) downward to the LOW position.

5. Turn key (3) to the ON position. The indicator for the engine preheater will illuminate. Leave the key in the ON position until the indicator turns OFF.

Note: Always use the engine preheater before starting the engine.

6. Crank the engine by turning key (3) to the START position. When the engine starts, release the key.

Note: If the engine does not start, return key (3) to the OFF position before you try to start the engine again.

i03649909

Engine and Machine Warm-Up

SMCS Code: 1000; 7000

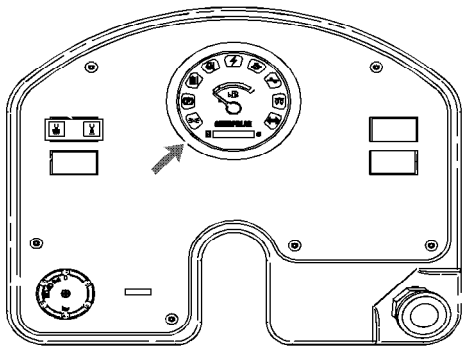


Illustration 45 g01032181

NOTICE
Keep the engine speed low until the engine oil indicator light goes out.

If the light does not go out within ten seconds, stop the engine and investigate the cause before starting again. Failure to do so, can cause engine damage.

1. Allow a cold engine to warm up at LOW idle for at least five minutes.

2. Look at the gauges and the indicator lights frequently during operation.

If the alternator indicator lamp remains illuminated after going to high idle, inspect the machine for the cause of the trouble. If the gauges do not respond properly, inspect the machine for the trouble. Before you move the machine, repair all problems.

3. Cycle all controls in order to allow warm oil to circulate through all of the lines and the cylinders.

Observe the following recommendations during the warm-up period for the engine:

- In temperatures above 0°C (32°F), the warm-up period is 15 minutes.
- In temperatures below 0°C (32°F), the warm-up period is 30 minutes or a longer period of time.
- In temperatures below −18°C (0°F), more time is required if the hydraulic controls are sluggish.

Parking

i01977821

Stopping the Machine

SMCS Code: 7000

NOTICE
Park the machine on a level surface. If it is necessary to park on a grade, securely block the tires and the drum.

Do not apply the parking brake while the machine is moving unless an emergency exists.

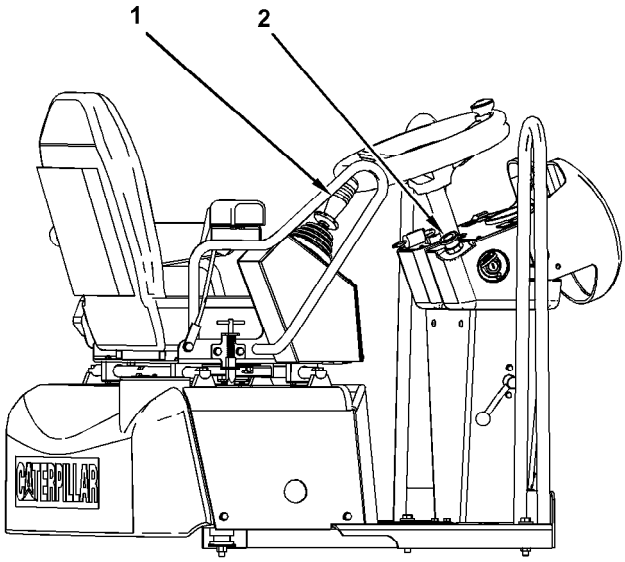


Illustration 46 g01032185

1. Move the propel control lever to the NEUTRAL position.
2. Apply the parking brake.

i02012285

Stopping the Engine

SMCS Code: 1000; 7000

NOTICE
Stopping the engine immediately after it has been working under load can result in overheating and accelerated wear of the engine components.

1. Before stopping the engine, allow the engine to run at low idle with no load for five minutes. This procedure allows the hot areas of the engine to cool gradually and the procedure will extend the life of the engine.
2. Turn the engine start switch to the OFF position. Remove the key.
3. Be sure that all of the controls are in the OFF position. Be sure that all of the controls are in the proper position for parking the machine.

i01977822

Stopping the Engine if an Electrical Malfunction Occurs

SMCS Code: 1000; 7000

Turn the engine start switch key to the OFF position. If the engine does not stop, perform the following procedure.

1. Open the engine access cover. Refer to the Operation and Maintenance Manual, “Access Doors and Covers” for more information.

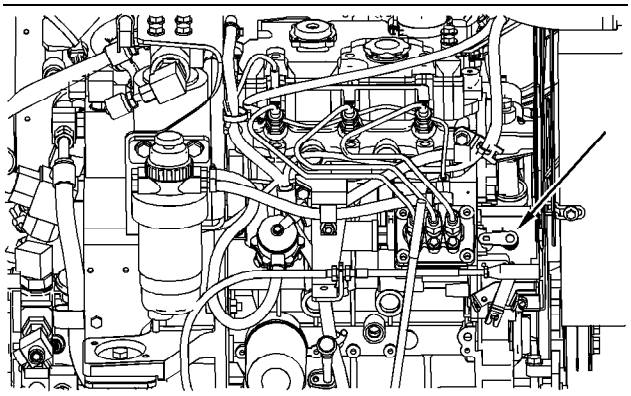


Illustration 47 g01032287

2. Rotate the lever that is on top of the fuel pump toward the engine. Hold the lever into this position until the engine stops running.

Note: Do not operate the machine until the cause of the problem is corrected.

i01994616

Leaving the Machine

SMCS Code: 7000

1. Use the steps and the handholds in order to dismount the machine. Face the machine in order to dismount the machine.

2. If the machine is being parked for an extended period of time, turn the disconnect switch for the battery to the OFF position.

3. Close all access covers and doors.

Vandalism Guard

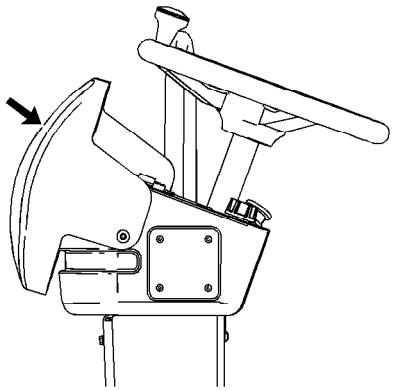


Illustration 48 g01041267

Vandalism guard in the open position

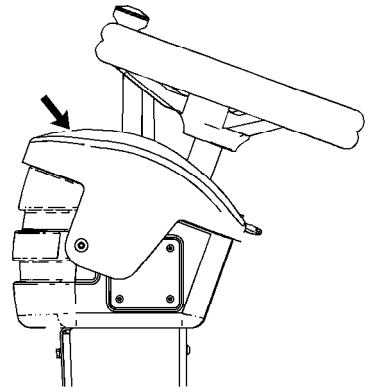


Illustration 49 g01041265

Vandalism guard in the closed position

Close and lock the vandalism guard when you leave the machine.

Transportation Information

i02202357

Shipping the Machine

SMCS Code: 7000; 7500

Investigate the travel route for overpass clearances. Make sure that there is adequate clearance for the machine that is being transported.

Remove ice, snow, or other slippery material from the loading dock and from the truck bed before you load the machine onto the transport machine. Removing ice, snow, or other slippery material will help to prevent the machine from slipping in transit.

NOTICE
Obey all laws that govern the characteristics of a load (height, weight, width, and length). Observe all regulations that govern wide loads.

When you move the machine to a colder climate, make sure that the cooling system has the proper antifreeze.

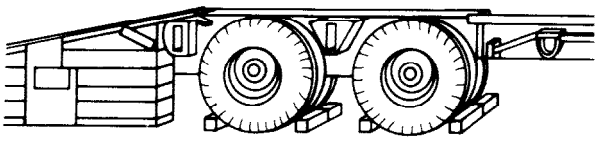


Illustration 50 g00303463

1. Before you load the machine, chock the trailer wheels or the rail car wheels, as shown.

2. Move the machine into position and engage the parking brake.

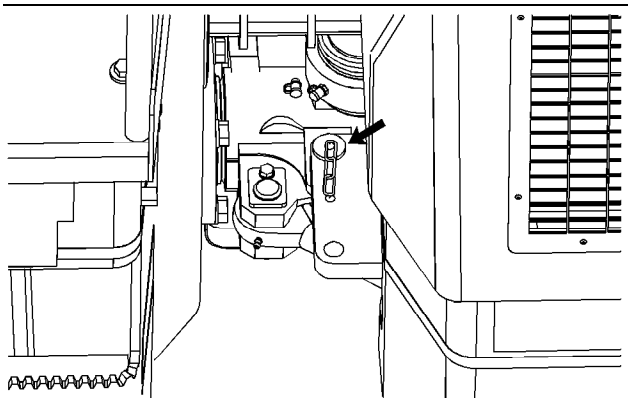


Illustration 51 g01031425

3. Stop the engine and install the steering frame lock into the LOCKED position. The pin will hold the front frame and the rear frame rigid.

Refer to the Operation and Maintenance Manual, “Steering Frame Lock” for further information.

4. Block the machine, and tie down the machine. Refer to the Operation and Maintenance Manual, “Lifting and Tying Down the Machine” for more information.

5. If your machine is equipped with a foldable rollover protective structure (ROPS), lower the ROPS. Refer to the Operation and Maintenance Manual, “Rollover Protective Structure (ROPS)” for further information on the ROPS.

6. Lock the doors and the access covers. Attach any vandalism protection. Install the cover on the console. Cover the operator seat.

7. To protect the cooling systems, mix the solution of antifreeze and water. The solution should provide protection to the lowest expected outside temperature. Drain the excess coolant into a suitable container.

8. Perform a walk-around inspection and measure the fluid levels in the various compartments.

9. Travel at a moderate speed. Observe all speed limitations when you are roading the machine.

Consult your Caterpillar dealer for shipping instructions for your machine.

i02854698

Roading the Machine

SMCS Code: 7000; 7500

S/N: 2251–Up

Note: Before roading the machine, consult your tire dealer for recommended tire pressures and speed limitations.

Inflate the tires to the correct pressure. Refer to the Operation and Maintenance Manual, “Tire Inflation Pressure Adjustment”.

The performance of the tires on the machine are specified by a rating of ton-kilometer per hour (ton-mile per hour). Consult your tire dealer for the specification of the tires on your machine. Observe the specified ton-kilometer per hour (ton-mile per hour) of the tires.

Perform a walk around inspection of the machine. Measure the fluid levels in the various compartments.

Check with the proper officials in order to obtain the required licenses.

Install any required flags, signals, or lights.

Travel at a moderate speed. Observe all speed limitations when you road the machine.

When you travel a long distance at full speed, stop the machine after 30 minutes.

In order to allow the tires to cool, stop the machine for 30 minutes.

In order to allow the components to cool, stop the machine for 30 minutes.

i01977826

Lifting and Tying Down the Machine

SMCS Code: 7000; 7500

NOTICE
Improper lifting or tiedowns can allow load to shift and can cause injury and damage.

For the basic specifications of the machine, refer to the Operation and Maintenance Manual, “Specifications”.

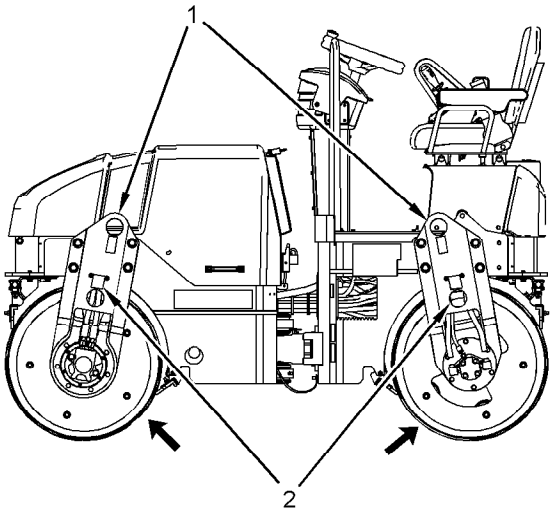


Illustration 52 g01032436

Lifting Point (1) – In order to lift the machine, attach the lifting devices to the lifting points.

Tie Down Point (2) – In order to tie down the machine, attach the tie-downs to the tie down points.

NOTICE
Do not tie down the machine over the operators platform. This will reduce the life of the drum and the platform isolation mounts.

Do not tie down the mahcine over the steering hitch area. This will reduce the life of the steering articulation bearings.

Use properly rated cables and properly rated slings in order to lift the machine.

Position the crane or the lifting device in order to lift the machine in a level position.

The width of the spreader bar must be sufficient to prevent the lifting cables or the lifting straps from contacting the machine.

1. Attach lifting cables to the lifting eyes. The lifting eyes are identified on the machine by symbols.

NOTICE
In all cases, before tying down the machine, place wooden blocks under the front and rear frames. This will reduce the stress on the rubber isolation blocks.

2. Place blocks under the front and the rear frames.

3. Place blocks under the front and rear drums. Block tires.

4. In order to secure the machine, use tie-downs during transportation.

Contact your Caterpillar dealer for shipping instructions.

Towing Information

i01977830

Towing the Machine

SMCS Code: 7000

⚠ WARNING

Improper hookup and towing is dangerous and could result in injury or death to yourself or others.

The towing connection must be rigid, or towing must be done by two machines of the same size or larger than the towed machine. Connect a machine on each end of the towed machine.

Be sure that all necessary repairs and adjustments have been made before a machine that has been towed to a service area, is put back into operation.

These towing instructions are for moving a disabled machine for a short distance at low speed. Move the machine at a speed of 2 km/h (1.2 mph) or less to a convenient location for repair. These instructions are only for emergencies. Always haul the machine if long distance moving is required.

Shielding must be provided on both machines. This will protect the operator if the tow line or the tow bar breaks.

Do not allow an operator to be on the machine that is being towed unless the operator can control the steering and/or the braking.

Before towing, make sure that the tow line or the tow bar is in good condition. Make sure that the tow line or the tow bar has enough strength for the towing procedure that is involved. The strength of the towing line or of the tow bar should be at least 150 percent of the gross weight of the towing machine. This is true for a disabled machine that is stuck in the mud and for towing on a grade.

Keep the tow line angle to a minimum. Do not exceed a 30 degree angle from the straight ahead position.

Quick machine movement could overload the tow line or the tow bar. This could cause the tow line or the tow bar to break. Gradual, steady machine movement will be more effective.

Normally, the towing machine should be as large as the disabled machine. Make sure that the towing machine has enough brake capacity, enough weight, and enough power. The towing machine must be able to control both machines for the grade that is involved and for the distance that is involved.

You must provide sufficient control and sufficient braking when you are moving a disabled machine downhill. This may require a larger towing machine or additional machines that are connected to the rear. This will prevent the machine from rolling away out of control.

All situation requirements cannot be listed. Minimal towing machine capacity is required on smooth, level surfaces. On inclines in poor condition or on surfaces in poor condition, maximum towing machine capacity is required.

Attach the towing device and the machine before you release the brakes.

Consult your Caterpillar dealer for towing a disabled machine.

Running Engine

If the engine is running, the machine can be towed for a short distance under certain conditions. The power train and the steering system must be operable.

The operator must steer the machine that is towed in the direction of the tow line.

Ensure that all instructions in this section are followed carefully. Ensure that all instructions in this section are followed exactly.

Stopped Engine

⚠ WARNING

Shutting off the engine will result in the loss of machine steering.

When the engine is stopped, additional steps may be required before the machine is towed. In order to avoid damaging the power train, the steering system, and the brakes, which may be inoperable, additional steps may be required.

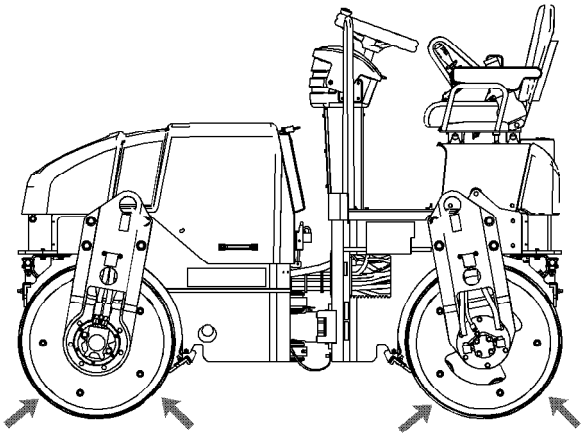


Illustration 53 g01032481

1. Block the drum or block the tires securely in order to prevent the movement of the machine. Do not remove the blocking until the tow vehicle has been positioned and the tow lines are in place.

2. Manually release the parking brake. Refer to the Operation and Maintenance Manual, “Parking Brake Manual Release” for more information.

NOTICE
Release the parking brake to prevent excessive wear and damage to the braking system when towing.

The procedure for manual release of the parking brake is outlined in the Operation and Maintenance Manual, “Parking Brake Manual Release”.

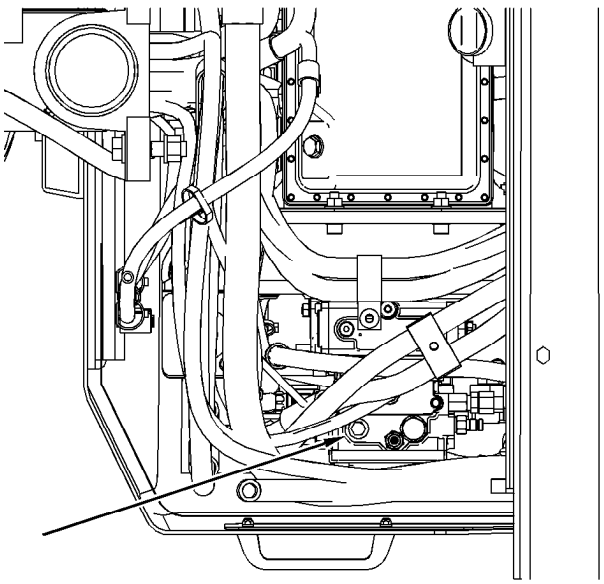


Illustration 54 g01032539

3. Turn the bypass valve for two full turns in the counterclockwise direction.

Note: Do not turn the bypass valve further than two turns. When the bypass valve is turned further than two turns, oil will leak past the bypass valve.

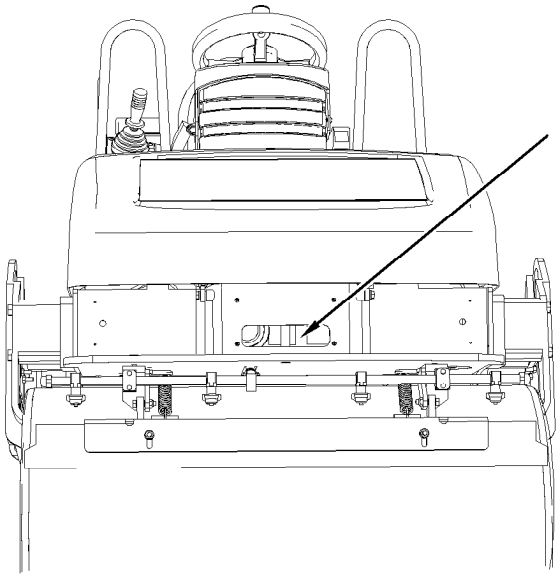


Illustration 55 g01032540
Front of the machine

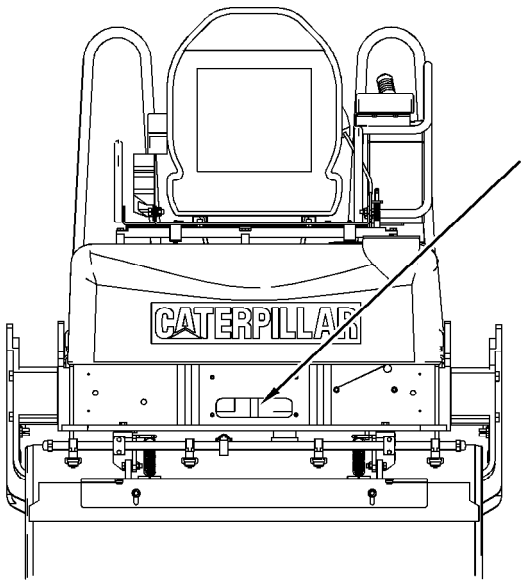


Illustration 56 g01032542
Rear of the machine

4. Attach the tow line to the machine at the tow points.

5. Attach the tow line to the vehicle that is used to tow the disabled machine.

6. Remove blocks from the drum or from the tires.

7. Tow the disabled vehicle at a slow rate of speed to the desired location.

8. Once the machine is at the desired location, securely block the drum or block the tires. This will prevent movement of the machine.

9. Turn the bypass valve clockwise in order to tighten the bypass valve.

10. Engage the parking brake.

11. Detach the tow lines.

i02816926

Parking Brake Manual Release

SMCS Code: 4267; 4354

⚠ WARNING

Personal injury or death can result from a brake malfunction. Do not operate the machine if the brake was applied due to a malfunction of the brake system.

Correct any problem before attempting to operate the machine.

There are two parking brakes on the CB-214E and the CB-224E machines. There is a parking brake in the front drum propel motor and there is a parking brake in the rear drum propel motor. The CB-225E has a parking brake in the front drum propel motor and a parking brake on each of the rear tire propel motors.

To manually release the parking brake on the drum propel motors, perform the following procedure:

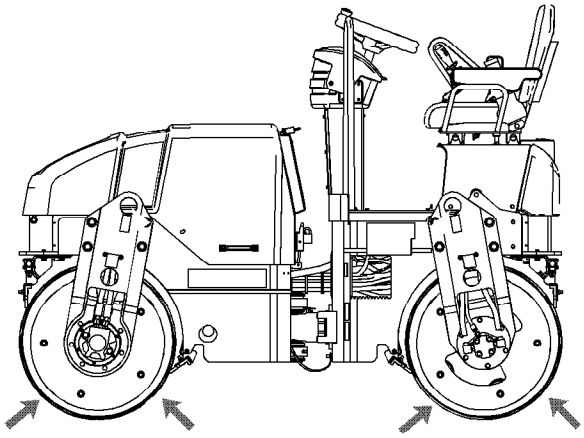


Illustration 57 g01032481

1. In order to prevent the machine from moving, block the drum and block the tires securely.

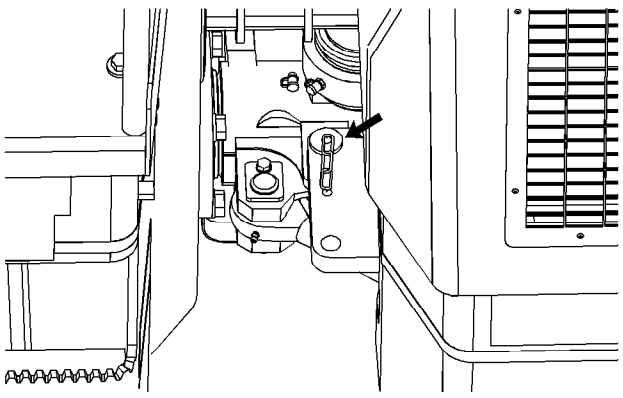


Illustration 58 g01031425

2. In order to hold the front frame and the rear frame rigid, install the steering frame lock into the LOCKED position.

Note: Refer to the Operation and Maintenance Manual, “Steering Frame Lock” for further information.

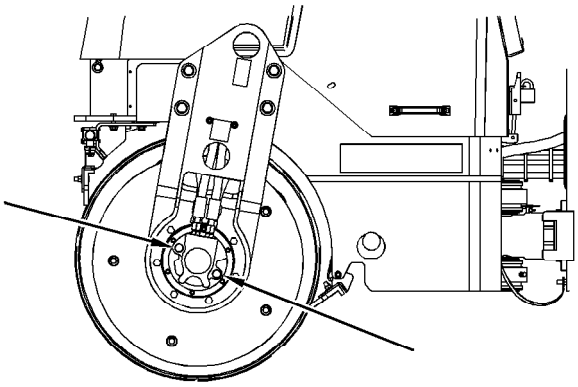


Illustration 59 g01032484

3. Remove the plugs in order to access the release screws for the front drum propel motor.

4. Evenly tighten the two release screws in order to compress the brake springs by approximately two turns.

Note: For the CB-214E and CB-224E, repeat step 3 and step 4, in order to manually release the parking brake on the rear drum propel motor.

Note: In order to manually release the parking brake on the rear tire propel motor on the CB-225E, complete step 5 and step 6. In order to manually release the parking brake on the rear drum propel motor on the CB-214E and CB-224E machines that are equipped with a manual brake release pump, complete step 5 and step 6.

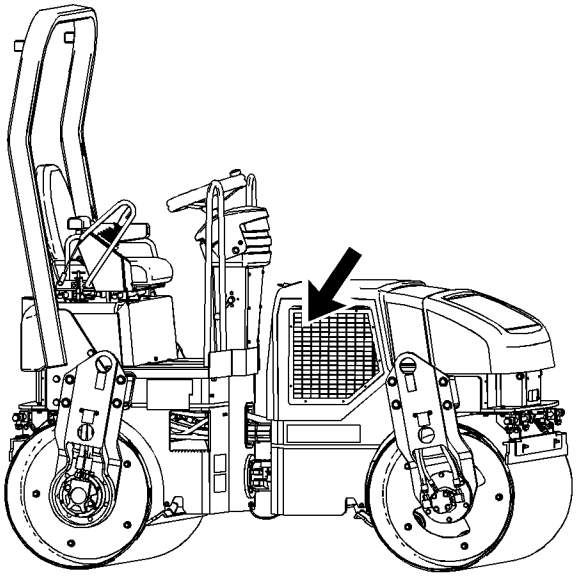


Illustration 60 g01389575

5. The manual brake release pump for the tire propel motor is located inside the engine compartment on the right side of the machine.

6. Open the valve and use the manual pump in order to release the parking brake for the tire propel motor.

Note: Close the valve before you operate the machine.

Engine Starting (Alternate Methods)

i03650200

Engine Starting with Jump Start Cables

SMCS Code: 1000; 7000

⚠ WARNING

Failure to properly service the batteries may cause peronal injury.

Prevent sparks near the batteries. They could cause vapors to explode. Do not allow the jump start cable ends to contact each other or the machine.

Do not smoke when checking battery electrolyte levels.

Electrolyte is an acid and can cause personal injury if it contacts skin or eyes.

Always wear eye protection when starting a machine with jump start cables.

Improper jump start procedures can cause an explosion resulting in personal injury.

Always connect the battery positive (+) to battery positive (+) and the battery negative (−) to battery negative (−).

Jump start only with an energy source with the same voltage as the stalled machine.

Turn off all lights and accessories on the stalled machine. Otherwise, they will operate when the energy source is connected.

NOTICE
When starting from another machine, make sure that the machines do not touch. This could prevent damage to engine bearings and electrical circuits.

Severely discharged maintenance free batteries do not fully recharge from the alternator after jump starting. The batteries must be charged to proper voltage with a battery charger. Many batteries thought to be unusable are still rechargeable.

This machine has a 12 volt starting system. Use only the same voltage for jump starting. Use of a welder or higher voltage damages the electrical system.

Refer to Special Instruction, Battery Test Procedure, SEHS7633, available from your Caterpillar dealer, for complete testing and charging information.

Use of Jump Start Cables

When the auxiliary start receptacles are not available, use the following procedure.

1. Make the initial determination of the machine's failure to crank. The procedure is applicable even if the machine does not have a diagnostic connector.
2. Move the propel control lever of the stalled machine into the NEUTRAL position. Engage the parking brake.
3. Turn the engine start switch on the stalled machine to the OFF position. Turn off all accessories.
4. Move the machine that is being used as a power source so that the jump start cables can reach the stalled machine. **DO NOT ALLOW THE MACHINES TO CONTACT EACH OTHER.**
5. Stop the engine on the machine that is being used as a power source. If you are using an auxiliary power source, turn off the charging system.
6. Check the battery caps for correct placement and for correct tightness. Make these checks on both machines. Make sure that the batteries in the stalled machine are not frozen. Check the batteries for low electrolyte.
7. Connect the positive jump start cable to the positive battery terminal on the stalled machine.

   Do not allow positive cable clamps to contact any metal except for the positive remote jump start terminal.
8. Connect the positive jump start cable to the positive terminal of the boost source.
9. Connect one end of the negative jump start cable to the negative terminal of the electrical source.
10. Make the final connection. Connect the negative cable to the frame of the stalled machine. Make this connection away from the battery, the fuel, the hydraulic lines, or moving parts.
11. Start the engine on the machine that is being used as a power source. If you are using an auxiliary power source, energize the charging system on the auxiliary power source.
12. Wait for a minimum of two minutes while the batteries in the stalled machine partially charge.
13. Attempt to start the stalled engine. Refer to Operation and Maintenance Manual, “Engine Starting”.